Iris Flower Classification Using Machine Learning

1. Introduction

The Iris Flower Classification project aims to develop a machine learning model capable of classifying Iris flowers into three species: Iris-setosa, Iris-versicolor, and Iris-virginica. The classification is based on four flower measurements: sepal length, sepal width, petal length, and petal width.

Machine learning classification algorithms can identify patterns in data and make accurate predictions for unseen samples.

2. Objective

To build a machine learning model that classifies Iris flowers into their respective species using sepal and petal measurements.

3. Dataset Description

The Iris dataset contains 150 flower samples and 5 columns.

Features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable:

- Species

Total Records: 150

4. Tools and Libraries Used

- Python
- Pandas

- NumPy
- Scikit-learn
- Matplotlib
- Google Colab

5. Methodology

Data Loading

The dataset was loaded using the Pandas library.

Data Preprocessing

- Checked dataset information
- Verified there were no missing values
- Separated features and target variable
- Encoded species labels into numerical values

Data Splitting

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

Model Training

A Random Forest Classifier was used to train the model.

Prediction

The trained model predicted flower species on the testing dataset.

Evaluation

Model performance was evaluated using Accuracy Score and Classification Report.

6. Results

Accuracy Achieved: 100%

Classification Report:

Precision: 1.00

Recall: 1.00

F1-Score: 1.00

The model successfully classified all test samples correctly.

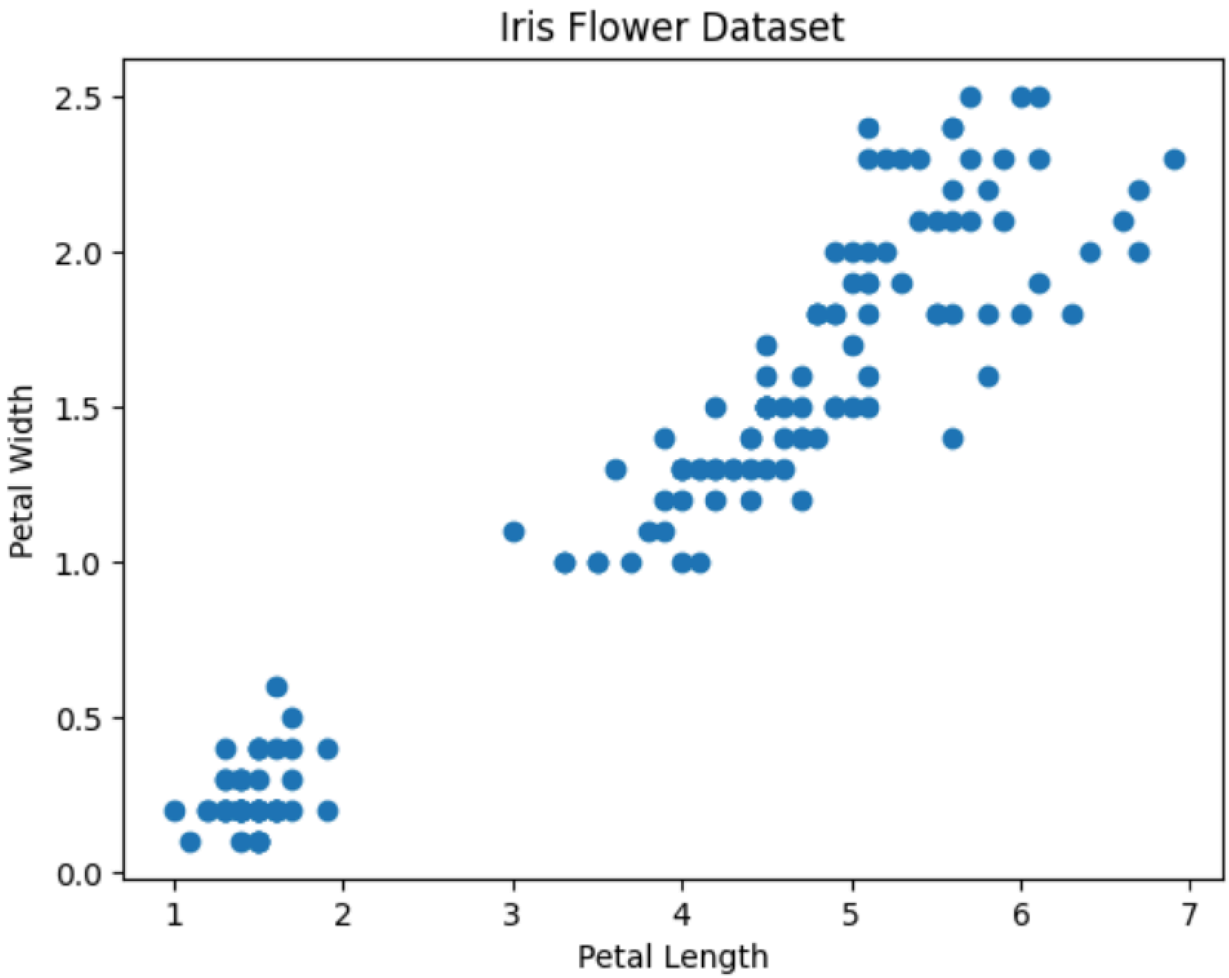
7. Visualization

[Insert Scatter Plot Screenshot Here]

The scatter plot visualizes the relationship between petal length and petal width in the Iris dataset.

8. Conclusion

The Iris Flower Classification model was successfully developed using the Random Forest algorithm. The model achieved 100% accuracy on the testing dataset and accurately classified Iris flowers into their respective species. This project demonstrates the effectiveness of machine learning techniques for classification problems.


Iris Flower Dataset
Petal Width
Petal Length
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7